

Chapter 18 Catastrophic Payment Results

OOPG and OOP, Nepal NLSS IV 2022/23

Arpan's calculations from NLSS IV

Purpose of This Chapter 18 Output

This report follows Chapter 18 of *Analyzing Health Equity Using Household Survey Data*. The chapter studies catastrophic health payments by combining three ideas: incidence, intensity, and distribution. Incidence asks whether a household crosses a catastrophic threshold. Intensity asks how far beyond the threshold the household goes. Distribution-sensitive measures ask whether the burden is concentrated among poorer or better-off households.

The report applies these ideas to two household out-of-pocket payment scopes. **OOPG** is out-of-pocket spending on communicable disease or injury from NLSS Section 8(B). **OOP** adds NCD out-of-pocket spending to OOPG after converting the NCD amount from annual to monthly terms. The two scopes are shown side by side because OOPG captures a narrower acute burden, while OOP captures a broader household health-payment burden.

The estimates use household survey weights. The primary total-expenditure denominator is reconstructed monthly nominal household consumption with OOPG/general health spending added back. The primary nonfood denominator is real monthly household nonfood consumption with OOPG added back. Distribution-sensitive measures rank households by pre-OOP welfare.

OOPG: Catastrophic Payment Results

OOPG refers to out-of-pocket spending on communicable disease or injury from NLSS Section 8(B). It is the narrower health-payment concept in this report, so it captures an acute communicable/injury-related burden rather than the full household health-payment burden. The purpose of this section is to read the Chapter 18 measures in sequence: first whether households cross the catastrophic threshold, then how far they cross it, and finally where that burden falls in the welfare distribution.

Incidence and intensity

Table 1 for OOPG answers two linked questions. The head count, H , asks how many households cross each threshold. The overshoot, O , asks how far the population lies above the threshold on average. The mean positive overshoot, MPO, then focuses only on households that crossed the threshold.

Catastrophic payment measure	5%	10%	15%	25%	40%
<i>OOPG as share of total expenditure</i>					
Head count (H)	15.06%	7.40%	4.10%	1.76%	—
standard error	0.67%	0.41%	0.28%	0.16%	—
Overshoot (O)	1.37%	0.84%	0.56%	0.29%	—
standard error	0.08%	0.06%	0.05%	0.03%	—
Mean positive overshoot (MPO)	9.08%	11.35%	13.74%	16.41%	—
<i>As share of nonfood expenditure</i>					
Head count (H)	—	—	9.73%	4.98%	2.16%
standard error	—	—	0.49%	0.32%	0.20%
Overshoot (O)	—	—	1.57%	0.87%	0.37%
standard error	—	—	0.10%	0.07%	0.04%
Mean positive overshoot (MPO)	—	—	16.17%	17.55%	17.09%

Table 1: Incidence and intensity of catastrophic OOPG payments.

At the 10% total-expenditure threshold, 7.40% of households experience catastrophic OOPG spending. At the stricter 40% capacity-to-pay threshold, the head count is 2.16%. The 10% total-expenditure overshoot is 0.84%, while the MPO is 11.35%. This means that among households crossing the 10% threshold, average spending is not just slightly above the threshold; it exceeds the threshold by the MPO amount.

The nonfood denominator gives a stricter view of financial burden because it is based on nonfood expenditure with health payments added back. For OOPG, the 15% total-expenditure head count is 4.10%, while the 15% nonfood-expenditure head count is 9.73%. The same health payment therefore appears more burdensome when measured against the household's nonfood capacity.

Budget-share curve

The budget-share curve is the visual counterpart to the incidence table. Households are ranked from left to right by decreasing health-payment budget share. The vertical axis shows the health-payment share. The upper curve uses nonfood expenditure as the denominator, and the lower curve uses total expenditure.

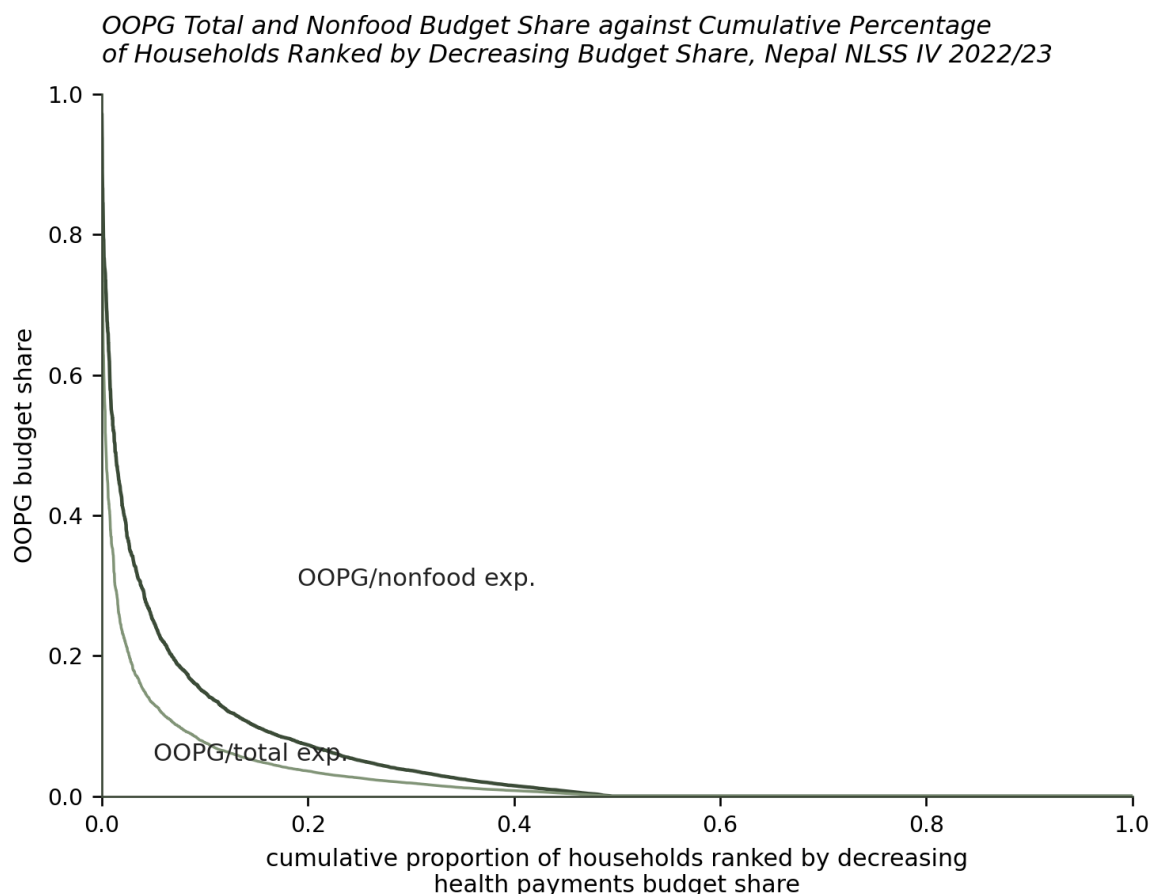


Figure 1: OOPG budget-share curve. Households are ranked by decreasing health-payment budget share.

The steep left-hand side of the curve shows that the highest budget shares are concentrated among a relatively small group of households. The nonfood curve lies above the total-expenditure curve because food spending is excluded from the nonfood denominator. This is why the nonfood head counts in the table are higher at comparable thresholds.

Distribution-sensitive measures

Chapter 18 then asks whether catastrophic payments are concentrated among poorer or better-off households. The concentration index is the key statistic here. A negative value means the burden is concentrated among poorer households, a positive value means it is concentrated among better-off households, and a value near zero means the burden is more evenly spread.

Distribution-sensitive measure	5%	10%	15%	25%	40%
<i>OOPG as share of total expenditure</i>					
Concentration index, C^H	-0.0210	0.0490	0.1221	0.2758	—
Rank-weighted head count, H^W	15.38%	7.04%	3.60%	1.28%	—
Concentration index, C^O	0.1595	0.2562	0.3427	0.4955	—
Rank-weighted overshoot, O^W	1.15%	0.62%	0.37%	0.15%	—
<i>As share of nonfood expenditure</i>					
Concentration index, C^H	—	—	-0.0320	0.0552	0.1797
Rank-weighted head count, H^W	—	—	10.05%	4.71%	1.77%
Concentration index, C^O	—	—	0.1329	0.2289	0.3825
Rank-weighted overshoot, O^W	—	—	1.37%	0.67%	0.23%

Table 2: Distribution-sensitive catastrophic payment measures. Households are ranked by pre-OOP welfare.

For OOPG, the concentration index for the 5% total-expenditure head count is -0.0210, which is close to zero. At the 25% total-expenditure threshold, it rises to 0.2758. This pattern means that higher-threshold catastrophic payments are more concentrated among better-off households. This should not be interpreted as poor households being protected. It may instead reflect lower ability among poor households to obtain and pay for expensive treatment.

The rank-weighted head count adjusts the ordinary head count for this distributional pattern. At the 10% total-expenditure threshold, the rank-weighted head count is 7.04%. Under the nonfood denominator, the 15% concentration index is -0.0320, showing how the equity interpretation can change when the denominator focuses on nonfood capacity rather than total expenditure.

OOP: Catastrophic Payment Results

OOP combines OOPG with NCD spending converted from annual to monthly terms. It is the broader health-payment concept in this report and therefore captures more of the household financial burden created by health care. The purpose of this section is to read the Chapter 18 measures in sequence: first whether households cross the catastrophic threshold, then how far they cross it, and finally where that burden falls in the welfare distribution.

Incidence and intensity

Table 1 for OOP answers two linked questions. The head count, H , asks how many households cross each threshold. The overshoot, O , asks how far the population lies above the threshold on average. The mean positive overshoot, MPO, then focuses only on households that crossed the threshold.

Catastrophic payment measure	5%	10%	15%	25%	40%
<i>OOP as share of total expenditure</i>					
Head count (H)	32.68%	18.47%	11.45%	5.63%	—
standard error	0.94%	0.64%	0.54%	0.33%	—
Overshoot (O)	4.35%	3.12%	2.39%	1.58%	—
standard error	0.22%	0.20%	0.18%	0.15%	—
Mean positive overshoot (MPO)	13.31%	16.90%	20.89%	28.13%	—
<i>As share of nonfood expenditure</i>					
Head count (H)	—	—	24.09%	14.27%	7.57%
standard error	—	—	0.74%	0.54%	0.40%
Overshoot (O)	—	—	7.02%	5.16%	3.57%
standard error	—	—	0.39%	0.35%	0.31%
Mean positive overshoot (MPO)	—	—	29.15%	36.12%	47.11%

Table 3: Incidence and intensity of catastrophic OOP payments.

At the 10% total-expenditure threshold, 18.47% of households experience catastrophic OOP spending. This is much larger than the corresponding OOPG estimate, showing how strongly NCD spending changes the picture. At the stricter 40% capacity-to-pay threshold, the head count is 7.57%. The 10% total-expenditure overshoot is 3.12%, while the MPO is 16.90%. This means that among households crossing the 10% threshold, average spending is not just slightly above the threshold; it exceeds the threshold by the MPO amount.

The nonfood denominator gives a stricter view of financial burden because it is based on nonfood expenditure with health payments added back. For OOP, the 15% total-expenditure head count is 11.45%, while the 15% nonfood-expenditure head count is 24.09%. The same health payment therefore appears more burdensome when measured against the household's nonfood capacity.

Budget-share curve

The budget-share curve is the visual counterpart to the incidence table. Households are ranked from left to right by decreasing health-payment budget share. The vertical axis shows the health-payment share. The upper curve uses nonfood expenditure as the denominator, and the lower curve uses total expenditure.

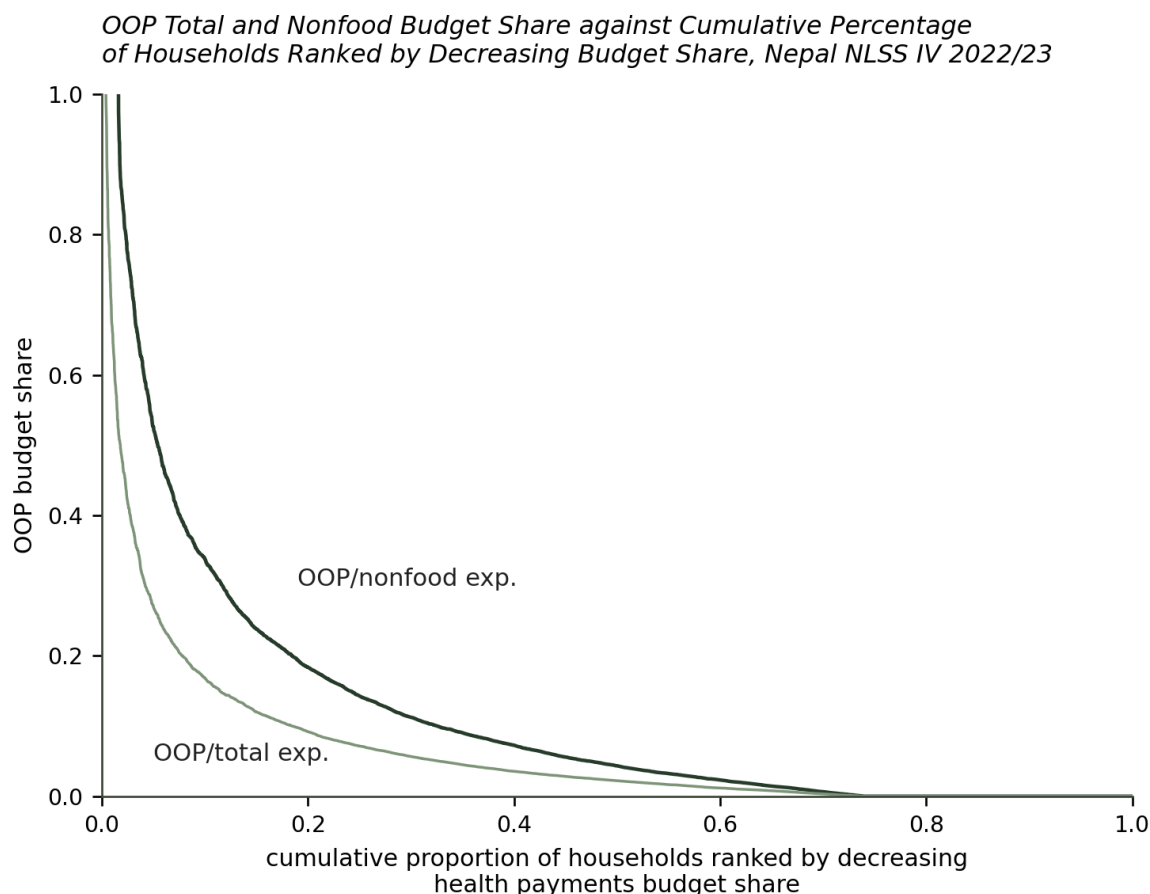


Figure 2: OOP budget-share curve. Households are ranked by decreasing health-payment budget share.

The steep left-hand side of the curve shows that the highest budget shares are concentrated among a relatively small group of households. The nonfood curve lies above the total-expenditure curve because food spending is excluded from the nonfood denominator. This is why the nonfood head counts in the table are higher at comparable thresholds.

Distribution-sensitive measures

Chapter 18 then asks whether catastrophic payments are concentrated among poorer or better-off households. The concentration index is the key statistic here. A negative value means the burden is concentrated among poorer households, a positive value means it is concentrated among better-off households, and a value near zero means the burden is more evenly spread.

Distribution-sensitive measure	5%	10%	15%	25%	40%
<i>OOP as share of total expenditure</i>					
Concentration index, C^H	0.0179	0.0781	0.1167	0.2171	—
Rank-weighted head count, H^W	32.10%	17.03%	10.11%	4.41%	—
Concentration index, C^O	0.2354	0.3102	0.3769	0.4884	—
Rank-weighted overshoot, O^W	3.33%	2.15%	1.49%	0.81%	—
<i>As share of nonfood expenditure</i>					
Concentration index, C^H	—	—	0.0093	0.0662	0.1552
Rank-weighted head count, H^W	—	—	23.86%	13.33%	6.40%
Concentration index, C^O	—	—	0.2162	0.2821	0.3606
Rank-weighted overshoot, O^W	—	—	5.50%	3.70%	2.28%

Table 4: Distribution-sensitive catastrophic payment measures. Households are ranked by pre-OOP welfare.

For OOP, the concentration index for the 5% total-expenditure head count is 0.0179, which is close to zero. At the 25% total-expenditure threshold, it rises to 0.2171. This pattern means that higher-threshold catastrophic payments are more concentrated among better-off households. This should not be interpreted as poor households being protected. It may instead reflect lower ability among poor households to obtain and pay for expensive treatment.

The rank-weighted head count adjusts the ordinary head count for this distributional pattern. At the 10% total-expenditure threshold, the rank-weighted head count is 17.03%. Under the nonfood denominator, the 15% concentration index is 0.0093, showing how the equity interpretation can change when the denominator focuses on nonfood capacity rather than total expenditure.

Overall Lesson

The central lesson is that catastrophic health spending is visible under both definitions, but the burden is much larger when NCD spending is included. OOPG identifies the communicable/injury-related part of the problem. OOP shows the broader financing burden faced by households. For the paper, OOP should carry the main headline result, while OOPG should be used as a narrower comparison that helps show how much of the burden is added when NCD payments are included.

The equity interpretation also requires care. Positive concentration indices at higher thresholds show that the most extreme measured payments are more common among better-off households. In a health-financing setting, this can reflect greater ability to seek and pay for care, not necessarily lower need among poor households. This point connects naturally to the next step, Chapter 19, where we study whether health payments are associated with impoverishment.